The Commutator of Representation

The electron, dealt from the throw and never from the lift

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October 2026

The fifth card. It is the one the dealer talks over, and it is never turned face up.

“Does the mechanism demonstrating this proof look like an electron to you?”

Reading note

This card is the patter: what the dealer says while the cards move.

The object. A card is carried, transformed, copied, projected and read again. At every move some of it comes through and some of it does not. The carrier carries; the reader reads; the throw throws. What a record keeps under the moves of the game is what this card is about, and nothing else.

The rule. **This card never lifts a card.** A reader sends many states to one reading. The states it sends to one reading are a fibre, and a fibre is what the record cannot resolve unless a receipt was kept with it. To lift a card is to name a point of a fibre that no receipt resolves. Every line calibrated *proved here* is a statement about readings and receipts and holds whichever point of the fibre lies under the card. The numerical card prints one such point, its certified reading. This card quotes it, with its source, where the electron is dealt (Section 7), and no proved line depends on its digits.

The destination. A carrier whose relation to the electron can be examined: a state that carries a phase, a rule for comparing two of them, the transport the rule requires, the curvature the transport cannot avoid, the commutator that measures the curvature, and the action whose stationary system the numerical card derives. The chain stops one step short, at the coupling, and the step is the knower's (Section 7).

Each statement carries one of five calibrations. *Proved here*: the proof is on this card. *Standard*: cited. *Measured*: a count or a definition taken from the declarations of the middle card, which is named only in Appendix A; it is never an experiment unless an experiment is cited. *Recorded*: a written chain of steps whose every premise is a cited result or a definition of this card; the steps are a record, and that the record is mathematics is the reader's reading of it. *Proposed*: a continuation the knower carries, stated and not derived, calibrated so that it is never mistaken for a step. The press check is Section 9: two readings of one register at two counts.

In five lines

1. A card carries a slot that no decision queries. A mark fits any carrier. *Proved here* (Section 1).

2. The throw moves cards, not the quarter. What survives is what the length family agrees on, and what a receipt retains. *Proved here* (Section 2).

3. Once around, the vectors return and the spinor has not. Twice around, the spinor returns. Clock and clock complement, at four and at eight. *Proved here* (Section 3); that the two eights are one eight is *proposed*.

4. Carry the phase and compare it, and the comparison is a connection. Go around one way and then the other, and the commutator is the curvature. *Proved here* (Section 6).

5. The electron is the carrier, dealt with the device's coupling. One step is not taken: that the device's coupling is the laboratory's. *Proposed* (Section 7).

1 The table: carriers, marks, tapes, readers

The objects of the game are defined here as mathematics, with no reference to any construction. Where a definition restates one of the middle card's, that is said and calibrated *measured*; the definition stands on its own.

Definition 1.1 (Record, continuation)**.** A *record* is a finite sequence of symbols from a finite alphabet. Its *length* is the number of symbols. The *continuation* of a record with at least two symbols is the record with its first symbol removed. A *record map* is a map from records to records.

Definition 1.2 (Carrier)**.** A *carrier* is a symbol s, a nonempty record q called the *slot*, and a record map ι, the *interaction*, such that

(i) for every record x of length at least two, $\iota(x)$ is the first symbol of x followed by q, so that the continuation of $\iota(x)$ is q;

(ii) for every record x of length one, $\iota(x)$ is the one-symbol record of the truth symbol.

The slot is *carried but not queried*: which clause applies to x depends only on the length of x, the symbol ι keeps depends only on x, and q is transmitted unchanged into every continuation. The map ι writes q, and so, as a function of the carrier, depends on q; no decision of ι, and no other datum of the carrier, is computed from q.

Measured: this is the middle card's carrier, restated under the encoding of Appendix A. The slot is the quarter under the card. In the game, the quarter is glued to the table; in the definition, the slot is written into every continuation and queried by no decision.

Definition 1.3 (Mark)**.** A *mark* is a proposition together with a decision of it: a fact that *clicks* (decides true) or does not. The *click* of a mark is the Boolean its decision returns.

A mark is the unit of distinction: two outcomes are told apart by a mark that clicks on one and not the other.

Definition 1.4 (Tape; three readers)**.** A *tape* is a finite list of marks. Three readers take a tape of length n with click bits $b_1, \dots, b_n$ to a natural number:

$$\begin{aligned} \text{the tally} \quad & T(t) = n, \\ \text{the register} \quad & R(t) = \sum_{i=1}^{n} b_i \, 2^{\,n-i} \quad \text{(a fold: } a \mapsto 2a + b_i \text{, starting from 0)}, \\ \text{the click count} \quad & C(t) = \#\{i : b_i = 1\}. \end{aligned}$$

Proposition 1.5 (Three cards, one tape)**.** *For a tape of length n: $R(t) = 2^n - 1$ if and only if every mark clicks; $C(t) = n$ if and only if every mark clicks; and in that case $R(t) + 1 = 2^{T(t)}$. So the three faces agree, in the sense that the register is the monomial of the tally and the click count is the tally, exactly when every mark on the tape holds.*

Proved here.

Proof. $R(t) \le \sum_i 2^{n-i} = 2^n - 1$ with equality if and only if every $b_i = 1$. $C(t) \le n$ with equality under the same condition. The identity $R + 1 = 2^n$ is the geometric sum. □

The register's reading of n unit cells is $2^n - 1$: a positional reading, which the bent card calls the second arithmetic (its Proposition 1.4) and which adds the fencepost to a tally to make a monomial. The three readers are the three cards face down. They show one face only when every affixed fact holds.

Definition 1.6 (Fibres of a reader; the length family)**.** A *reader* is a map F from tapes to a set. Its *kernel* is the partition of the tapes into the fibres $F^{-1}(v)$: what the reader does not see. A reader F *refines* a reader G when every fibre of F lies in a fibre of G, equivalently when $G = g \circ F$ for some map g. The *length family* Λ is the set of readers that factor through the tally, $G = g \circ T$. On this card "every reader" means every member of Λ, and nothing larger. A *payload reader* is a reader not in Λ: one that separates two tapes of equal length.

Proposition 1.7 (What each reader distinguishes)**.** *(a) The fibre of T at n is the set of all tapes of length n: the tally forgets every proposition and every click.*

(b) The pair (T, R) separates click sequences: two tapes have equal tally and equal register if and only if their click sequences are equal. The register alone does not: a non-clicking mark followed by a clicking one, and a single clicking mark, both read 1*.*

(c) For $n \geq 1$ the click count is a payload reader: two tapes of length one, one clicking and one not, have equal tally and different click count. It retains one Boolean distinction per cell, summed.

(d) No reader of click bits separates two tapes whose marks click alike but carry different propositions. A receipt naming the propositions (Definition 2.4) refines the click sequence to recovery.

Proved here.

Proof. (a) is the definition of T. (b) Given n, the integer $R \in [0, 2^n - 1]$ has exactly one expansion in n binary digits, leading zeros included, and those digits are $b_1, \ldots, b_n$. For the counterexample, the two-cell tape reads $0 \cdot 2 + 1 = 1$ and the one-cell tape reads 1. (c) and (d) are displayed by the examples. □

So the tally forgets the payload; the click count keeps a Boolean distinction; the register, read with its length, keeps order and bits; and a receipt refines an observation until it recovers. Four readers, four kernels, one tape.

Proposition 1.8 (A mark fits any carrier)**.** *Suppose no field of a mark refers to its carrier. Then for any two carriers X and Y the map that copies each field is a bijection from marks on X to marks on Y, and a mark and its copy agree in every field. In particular a mark placed on a carrier of electric charge and the same mark placed on an empty carrier agree in every field.*

Proved here; it is the cheat sheet's Proposition 6.1.

Proof. The copy is the identity on each field, and by hypothesis no field changes when the carrier does. □

This is universality bought by blindness. The mark fits the carrier of charge exactly as well as it fits a carrier with nothing in it, because it never looks at the carrier at all.

Proposition 1.9 (Two readers agree exactly when the lengths agree)**.** *Let U be the reader that keeps the length of a tape and writes the truth symbol in every cell. If every cell of a tape s is the truth symbol, then for every tape u, $s = U(u)$ if and only if $T(s) = T(u)$.*

Proved here; it is the cheat sheet's Proposition 6.2.

Proof. $U(u)$ is the tape of length $T(u)$ whose every cell is the truth symbol, by induction on u; two such tapes are equal if and only if their lengths are. □

Proposition 1.10 (Re-lettering is free)**.** *Let ρ overwrite every cell of a tape with the truth symbol. Any reader F that factors through the tally, $F = g \circ T$, satisfies $F(\rho\, t) = F(t)$ for every tape t.*

Proved here.

Proof. $T(\rho\, t) = T(t)$ since ρ changes no length. □

Measured: the middle card's own reading of its tapes factors through the tally, and overwriting every cell of every tape with the one truth symbol leaves its pair of readings fixed.

Overwriting is free, and it is not a symmetry. Four different things are called by the one word, and they are separated here before any of them is read as a charge.

Proposition 1.11 (Relabelling, erasure, conservation, Noether)**.** *Let the alphabet have at least two symbols.*

(1) A relabelling *applies a bijection π of the alphabet to every cell. Relabellings are invertible and form a group isomorphic to the permutations of the alphabet; each preserves T and every reader in Λ.*

(2) The erasure *ρ of Proposition 1.10 satisfies $\rho \circ \rho = \rho$ and is not injective on tapes of length at least one. It is therefore not an element of any group of maps of tapes, and it preserves every reader in Λ.*

(3) A quantity Q is conserved *by an evolution E of tapes when $Q \circ E = Q$. The tally is conserved by every permutation of cells (a throw) and by ρ. It is not conserved by writing: if W appends one mark, $T \circ W = T + 1$.*

(4) A Noether current *requires an action on a space of fields and a continuous one-parameter group of transformations leaving it invariant; on solutions of the Euler–Lagrange equations the current then has vanishing divergence.*

None of (1)–(3) supplies (4): tapes form a discrete set, ρ belongs to no group, and no action on tapes is given.

Proved here for (1)–(3), by inspection of the maps; *standard* for (4) [18, 23]. The continuous group, the action and the current of (4) are supplied in Section 6, where the constant phase rotation of a spinor leaves the Maxwell–Dirac action invariant and its current is conserved (Proposition 6.9). *Proposed*: Noether in the game's words. The contents of the cells are the symmetry; the count of cells is the charge. The correspondence sends invariance under ρ to symmetry and the tally to charge. It preserves the factorization through the tally. It does not preserve invertibility, continuity, or conservation under writing, which is the one evolution the device performs.

Proposition 1.12 (Writing is not free)**.** *Appending one entry to a record costs one step, and a counter that increases by one per entry is the tally of those steps. Over the field with one element every map is the identity and counting costs nothing; writing does not.*

Recorded, from the numerical card's Remark 8.3 and Definition 1.4: the tally counts symbols, and each symbol was written in one step.

The throw, seen by the tally. A throw is a permutation of carriers on the table. A reader of lengths cannot see it: by Proposition 1.9 every tape of a given length reads the same, and by Proposition 1.10 the contents are invisible to any reader in the length family. The dealer's hands are invisible to the tally. That is why the mark loses: the mark reads motion, and motion is not on the tape.

2 The throw: what is preserved, and the rule

Proposition 2.1 (The box has one point)**.** *Compare pairs of natural numbers by $(a, b)\,\mathcal{R}\,(c, d)$ if and only if $ad \leq cb$. Any two pairs are related one way or the other, so the equivalence generated by $\mathcal{R}$ has exactly one class, and every function out of the quotient is constant.*

Proved here.

Proof. For any two pairs either $ad \leq cb$ or $cb \leq ad$, since the order on the natural numbers is total. Hence any two pairs are related by $\mathcal{R}$ or by its reverse, and the generated equivalence identifies them. A function on a one-point set is constant. □

The numerical card describes this as an erased channel (its Section 7.5): a rule that merges all inputs into one class reports one value. The one value is "true equals true."

Proposition 2.2 (One rule, four readers)**.** *Let X be a set, r a relation on it, and X/r the classes of the equivalence r generates. (a) If $r(a, b)$ holds exactly when $f(a) = f(b)$ for a map f, then X/r is in bijection with the image of f. (b) If r has a least element, or if any two elements are related one way or the other, then X/r has exactly one class. Hence evaluation at every input keeps every function; a single decision keeps two classes, clicked and not clicked; an order on truths with a least element keeps one; an order on ratios that relates any two keeps one.*

Proved here; it is the cheat sheet's Proposition 6.3.

Proof. (a) is the first isomorphism theorem for sets. For (b), $x \sim x_0 \sim y$ for any x, y, or $x \sim y$ directly. □

Definition 2.3 (Representative)**.** A *representative* of a class is one element of it, selected. A division that returns a number from the one-point quotient of Proposition 2.1 selects a representative: it picks a point of the fibre.

Definition 2.4 (Observation, receipt)**.** An observation on a state domain D is a map $O : D \to Y$; its fibre at y is $O^{-1}(y)$. A *receipt* is a map $\mathcal{R} : D \to A$ retained with the observation; it supports exact recovery when $(O, \mathcal{R})$ is injective, which holds exactly when $\mathcal{R}$ is injective on each fibre of

O. If a fibre has m states, a receipt alphabet supporting recovery on it has at least m symbols, $\lceil \log_2 m \rceil$ bits.

Recorded, from the geometric card's Definition 4.1 and Propositions 4.2 and 4.3.

The title of this card is a square. A state is moved; the observation is taken before or after; the question is whether the two routes agree.

Definition 2.5 (Commuting square; commutator of representation)**.** Let $O : D \to Y$ be an observation and $U : D \to D$ a transformation of states. A map $V : Y \to Y$ *represents* U on observations when

$$O \circ U = V \circ O.$$

The *commutator of representation* of (O, U) is the set of pairs (x, x') with $O(x) = O(x')$ and $O(Ux) \neq O(Ux')$: the pairs a single fibre of O sends to two fibres.

Proposition 2.6 (When the square closes)**.** *A map V with $O \circ U = V \circ O$ exists if and only if the commutator of representation of (O, U) is empty, that is, if and only if U sends each fibre of O into a fibre of O; V is then unique on the image of O. On tapes with $O = T$: a throw and the erasure ρ are represented by the identity; writing one mark is represented by $n \mapsto n + 1$; the* click filter *K, which deletes every non-clicking mark, is represented by no map, since $T(Kt) = C(t)$ and the two tapes of length one of Proposition 1.7(c) lie in one fibre of T and are sent to lengths* 1 *and* 0*.*

Proved here.

Proof. If V exists, $O(x) = O(x')$ gives $O(Ux) = V(O(x)) = V(O(x')) = O(Ux')$. Conversely, if every fibre goes into a fibre, $V(O(x)) = O(Ux)$ is well defined on the image, and forced there. The examples are computations of T. □

Two kinds of thing can happen to a square, and the card keeps them apart. The square can close while the map $U \mapsto V$ forgets: two transformations of states with one image on observations. That is the spinor of Section 3.3, whose sign is lost and whose square closes (Proposition 3.11). Or two transformations of states can fail to commute with each other, and the failure can itself be observed. That is transport, whose commutator is the curvature (Theorem 6.7). The first is a fibre; the second is a field.

Theorem 2.7 (A representative is a choice)**.** *Let a datum be carried by every record of a family only as a projection, and be determined by no record of the family from its own contents, the family being closed under finite extension. Then any procedure that supplies a value of the datum is not a procedure of the family: it selects one element of a set the family can name but cannot enumerate to the point of selection.*

Recorded, from the Quarter's Theorem 3.1 and its proof, which is the standard step that a selection procedure inside the system would make the value representable there. The middle card keeps both extensionality principles and refuses choice, the third ingredient of Diaconescu's theorem [7]; *measured*, from a sweep of its declarations reported in the Quarter.

Definition 2.8 (Three levels of data; the rule of this card)**.** Fix a state domain D, an observation O on it and a receipt $\mathcal{R}$ retained with O. A datum of a state is

(a) *on the state* if it is any function on D;

(b) *on the observed class* if it is constant on each fibre of O, equivalently a function on the image of O;

(c) *in the retained receipt* if it is constant on each fibre of $(O, \mathcal{R})$.

A line of this card is *admissible* for $(O, \mathcal{R})$ only if its truth is a datum of level (b) or (c): it is unchanged when a state is replaced by another with the same observation and the same receipt. A *representative* is a point of a fibre of $(O, \mathcal{R})$; naming it is a datum of level (a) only, and is not admissible.

The operations act at fixed levels. Re-lettering (Proposition 1.10) and the exchange of one reader for another in the length family act on states and fix every level-(b) datum of the tally. Replacing a representative by another acts inside a fibre of $(O, \mathcal{R})$ and fixes every datum of levels (b) and (c). Lengths and parities are level-(b) data of the tally; clicks are level-(b) data of the click reader; the degree of a polynomial is a level-(b) datum of the degree reader (Definition 3.6). Signs and retained bits are level-(c) data. They are admissible because they are retained, not because they can be recovered from a quotient: a function on a quotient that identifies every pair is constant (Proposition 2.1), so a sign is not a function on the one-point box, and the sign after one revolution is not a function on the pair $\{\psi, -\psi\}$, which lies in one fibre of the vector observation (Proposition 3.11). *Proved here*: each assignment of a level is the definition of the reader concerned.

Corollary 2.9 (Why the number never comes up in a proof)**.** *The numerical card's printed reading is a representative: the card says so, and its receipt is the pair of rational walls that bracket it. By Definition 2.8, no digit of that reading, no coefficient of the residual it solves, no wall, gain, seed or discriminant can occur in a line of this card calibrated* proved here. *Where the reading enters, it enters as a quoted value with its source and its calibration, as a parameter of a definition, and every proved line about the object so defined holds for every value of that parameter.*

Recorded. This is the reason at the level of the record. It is proved below at two further levels: what the length family sees (Section 1) and what the degree hierarchy sees (Proposition 3.8). The parameter is the coupling of Definition 7.1.

Proposition 2.10 (The sign convention)**.** *Index the slots of a formula by $k = 0, 1, 2, \ldots$. A* signature *assigns each slot a sign; a literal in a signed slot reads its variable if the sign is $+$ and its negation if the sign is $-$. Four signatures, with their signs on the first four slots:*

$$\textit{euclidean}\ (++{+}{+}), \quad \textit{negated}\ (----), \quad \textit{west}\ (+---), \quad \textit{east}\ (-+{+}{+}),$$

the last two repeating with period four. Let Φ be the conjunction of two clauses, the first over slots $0, 1$ and the second over slots $2, 3$, each clause the disjunction of the two variables a, b in its slots. Read over the four assignments $(a, b) \in \{TT, TF, FT, FF\}$, the four signatures give

euclidean	$[T, T, T, F]$	$a \vee b$
negated	$[F, T, T, T]$	$\neg a \vee \neg b$
west	$[F, T, F, T]$	$\neg b$
east	$[T, F, T, F]$	b

The conjunction of the euclidean and negated readings is exclusive or, $[F, T, T, F]$. The conjunction of the west and east readings is $b \wedge \neg b$, the empty theory.

Proved here, by the table.

Proof. Under the euclidean signature both clauses read $a \vee b$. Under the negated signature both read $\neg a \vee \neg b$. Under west, slots $0, 1, 2, 3$ carry $+, -, -, -$, so the clauses read $a \vee \neg b$ and $\neg a \vee \neg b$, whose conjunction is $\neg b$. Under east they read $\neg a \vee b$ and $a \vee b$, whose conjunction is b. Then $(a \vee b) \wedge (\neg a \vee \neg b)$ is exclusive or, and $\neg b \wedge b$ is unsatisfiable. □

Measured: the middle card's table of the four signatures is this one, and its step constraint is the exclusive or. *Standard*: the quaternion carries two of the four at once, its basis elements squaring to $(+---)$ while its norm is $(++++)$ [11]. *Proposed*: the two definite signatures are a constraint met exactly, a multiplier; the two indefinite ones are a constraint penalized at a finite signal speed; and two penalties do not compose. Exclusive or, which is the gate the verdict of Section 5 is decided by, is writable only by combining the two definite signatures.

Proposition 2.11 (Same difference)**.** *Compare two truths a, b by what they are, $a = b$; and compare the marks that carry them by how they click, that is by the decision each carries. The two 2×2 tables agree in every cell, and each is unchanged when turned over. All a table carries is which diagonal clicks: $T = T$ and $F = F$ do, $T = F$ and $F = T$ do not, and the two off-diagonal cells are equal as values.*

Proved here, by the four cells; it is the cheat sheet's Proposition 6.4.

3 The flop: once around, twice around

3.1 The mirror, with the coefficients removed

The cheat sheet works the mirror on one residual with its coefficients written out. Here the same facts are proved for every quadratic residual, and the coefficients never appear; by Definition 2.8 that is the only form in which they are admissible.

Let $r(y) = ay^2 + by + c$ with $a > 0$ and two real roots $\alpha > \beta$, and let $S(y) = \frac{a}{3}y^3 + \frac{b}{2}y^2 + cy$, so that $S' = r$. Write $c_0 = -b/(2a)$ for the vertex and $\rho(y) = 2c_0 - y$ for reflection about it; ρ is an involution whose only fixed point is c_0.

Proposition 3.1 (The residual is even, the action is odd)**.** *For every real h,*

$$r(c_0 + h) = r(c_0) + ah^2, \qquad S(c_0 + h) - S(c_0) = r(c_0)\, h + \tfrac{a}{3}h^3. \tag{3.1}$$

So $r \circ \rho = r$ and $S \circ \rho - S(c_0) = -(S - S(c_0))$: the reflection fixes the residual and reverses the action. It exchanges the two roots, $\alpha = c_0 + \sqrt{-r(c_0)/a}$ and $\beta = c_0 - \sqrt{-r(c_0)/a}$.

Proved here.

Proof. $r'(c_0) = 0$, so the expansion of r about c_0 has no linear term and its quadratic term is ah^2. Integrating from c_0 gives the second identity, which is odd in h. The roots solve $ah^2 = -r(c_0)$, and $r(c_0) < 0$ since r has two real roots and positive leading coefficient. □

Proposition 3.2 (The reading is a least action)**.** *The equation $r(y) = 0$ is the Euler–Lagrange equation of S: for a function of one variable, stationarity is $S'(y) = 0$, and $S' = r$. Both roots are stationary. $S''(\alpha) = r'(\alpha) = 2a(\alpha - c_0) > 0$, so α is a strict local minimum of S; $S''(\beta) < 0$, so β is a strict local maximum. The branch above the vertex is the side of the mirror on which the stationary point is a minimum. Moreover, for every y,*

$$r'(y)^2 = (b^2 - 4ac) + 4a\, r(y), \tag{3.2}$$

so at either root the first derivative squares to the discriminant.

Proved here.

Proof. The first claims are Proposition 3.1 and the definition of S. For (3.2), $(2ay + b)^2 = 4a^2y^2 + 4aby + b^2$ and $4a\, r(y) = 4a^2y^2 + 4aby + 4ac$; subtract. □

The mirror acts on cells as well as on residuals. A cell symmetric about its centre samples f at $y - h$, y, $y + h$ with weights $1, -2, 1$.

Proposition 3.3 (The first order a mirror cell does not cancel)**.** *Let f be four times continuously differentiable on $[y - h, y + h]$. Then*

$$f(y + h) - 2f(y) + f(y - h) = h^2 f''(y) + \tfrac{h^4}{12} f''''(\xi)$$

for some $\xi \in (y - h, y + h)$. The mirror cell cancels every odd power of h, reproduces the second derivative, and its first error is the fourth-derivative term; it is exact for every f with $f'''' = 0$, that is, for every cubic.

Standard (Taylor's theorem with the mean-value form of the remainder, applied to $h \mapsto f(y + h) + f(y - h)$, which is even in h).

The continuum counterpart is the beam. On $[a_0, b_0]$ with interpolation conditions $\varphi(x_j) = v_j$ at knots $a_0 = x_0 < \cdots < x_N = b_0$, the bending functional

$$E[\varphi] = \tfrac{1}{2} \int_{a_0}^{b_0} \varphi''(x)^2 \, \mathrm{d}x$$

is minimized over twice weakly differentiable interpolants by the natural cubic spline; its first variation is $\int \varphi'' \eta'' \, \mathrm{d}x = 0$ for every admissible η, whose Euler–Lagrange equation between knots is $\varphi'''' = 0$. *Standard* [13, 5]. The two statements meet at one correspondence, and it is supplied, not derived: the first order a mirror cell leaves uncancelled is the fourth; the functional whose stationary equation sets exactly that order to zero is the bending energy; and its solutions between knots are cubics, the degree of S. *Proposed.*

Proposition 3.4 (Newton's exact error law; exact up to acceleration)**.** *For $e = y - \alpha$ and $d = \alpha - \beta$, the Newton map $N(y) = y - r(y)/r'(y)$ satisfies*

$$N(y) - \alpha = \frac{e^2}{2e + d} = \frac{r''}{2} \frac{e^2}{r'(y)} \tag{3.3}$$

exactly, wherever $r'(y) \neq 0$. The second difference of the residual is constant: $r(y + h) - 2r(y) + r(y - h) = 2ah^2$ for all y, h.

Proved here.

Proof. $r(y) = a\,e(e+d)$ and $r'(y) = a(2e+d)$, so $N(y) - \alpha = e - e(e+d)/(2e+d) = e^2/(2e+d) = ae^2/r'(y)$, and $a = r''/2$. The second difference of a quadratic is twice its leading coefficient times h^2. □

The word "acceleration" needs a variable, a clock and an evolution law, and they are supplied once.

Definition 3.5 (The reading clock)**.** The *reading clock* is a real parameter s, and the *reading motion* is the uniform evolution $y(s) = y_0 + sh$ of the variable. The *acceleration* of the residual along it is $\ddot{\rho}(s)$, where $\rho(s) = r(y(s))$.

For a quadratic residual $\ddot{\rho}(s) = 2ah^2$ for every s, and the second difference at unit clock step, $\rho(s+1) - 2\rho(s) + \rho(s-1)$, equals it exactly (Proposition 3.4). *Proved here.* In these terms the Newton step is the first variation, exact for any residual with zero acceleration, and by (3.3) its whole error is the one term proportional to r'', which the step never evaluates. The variational solution is exact up to a constant acceleration. *Proposed*: in the game's words this is Kepler, an areal velocity applied once, a conserved first-order quantity, with no force law ever read; the force law is upstream.

3.2 The derivative hierarchy

An action with at most three independent-field factors in every term, with fixed coefficients and linear operators, has $D^4S = 0$; its residual $R = DS$ is at most quadratic, $D^3R = 0$; its linearization DR is linear in the perturbation; and D^2R is constant. For bounded polynomials of degree at most three on a Banach space, the Fréchet and Gateaux derivatives coincide, and the same holds for the residual and its derivative. The four presentations of the numerical card, first-order vacuum gravity, fixed-Hamiltonian Schrödinger evolution, incompressible flow with response fields, and bare electrodynamics on a fixed background, each have this hierarchy in their declared regular sectors. *Proved in the numerical card* (its Theorems 1.1 and 2.4 and Proposition 2.5); the first-order gravitational mechanism is Deser's [6], the electrodynamic presentation is Tong's [23].

An action can be read three ways, and the three readers have different fibres.

Definition 3.6 (Three readers of an action)**.** Let X be a real Banach space and S a bounded polynomial functional on X of degree at most three. The *degree reader* returns the degree of S. The *derivative reader* returns the map $S \mapsto DS$. A *selected-evaluation reader* returns the values $DS(x)$ at the points x of a selected set $\Sigma \subset X$; the *stationary reader* returns the set $Z(S) = \{x : DS(x) = 0\}$.

Proposition 3.7 (The three fibres)**.** *On $X = \mathbb{R}$ write $S_g(y) = gy^3$.*

(a) The fibre of the degree reader at three contains S_g for every $g \neq 0$: the degree does not determine a coefficient.

(b) $DS = DS'$ if and only if $S - S'$ is constant. The derivative reader forgets exactly the additive constant. For S_g: $DS_g = 3gy^2$, $D^2S_g = 6gy$, $D^3S_g = 6g$, so $g = D^3S_g/6$ is read from the derivative.

(c) The selected-evaluation reader at $\Sigma = \{0\}$ returns 0 for every S_g: its fibre contains them all.

(d) $Z(\lambda S) = Z(S)$ *for every* $\lambda \neq 0$*: the stationary reader cannot see an overall normalization. It does see a relative coupling: for* $S(y) = \frac{1}{2}y^2 + \frac{g}{3}y^3$ *with* $g \neq 0$*,* $Z(S) = \{0, -1/g\}$*.*

Proved here.

Proof. (a) and (c) by evaluation. (b) $D(S - S') = 0$ on the connected space X gives $S - S'$ constant, and conversely; the derivatives of gy^3 are computed. (d) $D(\lambda S) = \lambda DS$; and $DS = y + gy^2 = y(1 + gy)$. □

Proposition 3.8 (The degree forgets the field; the derivative keeps it)**.** *The shared hierarchy identifies the four presentations in their degrees. Their field spaces, boundary sectors, coefficients, couplings, initial data and solutions are not functions of the degree: a Lorentzian metric sector, a complex Hilbert space, a divergence-constrained velocity and a gauge-and-spinor structure each carry a sequence* (S, R, DR, D^2R) *with* $D^3R = 0$*, and the degree hierarchy does not recover which one it came from. The derivative reader does: the sequence contains* S *itself, and every coefficient of a nonconstant term of* S *is a function of* DS*. A reader restricted to the degree hierarchy cannot reconstruct a coupling; a reader of the derivative reads it.*

Proved here, from Proposition 3.7 and the numerical card's shared hierarchy; the numerical card closes its Section 2.5 with the same caution: "the theorem supplies an exact shared derivative hierarchy; comparing their readings requires the declared observation maps." In one direction the Fréchet derivative is a number, the remainder is exactly quadratic, and nothing is left abstract (the cheat sheet's Proposition 2.1, which holds for every quadratic residual by the expansion $r(y + h) = r(y) + r'(y)h + ah^2$).

So the coupling survives the derivative. It sits in the interaction law, a coefficient of DS, and the derivative carries it everywhere the law goes. What forgets the coupling is the degree. Whether the electron of this card reads its law through its degree or through its derivative is a part of its construction, stated in Definition 7.1: it obeys a stationary system, so it reads the derivative, and its coupling is in its law.

3.3 The slack and the spinor

For $n \geq 1$ let L_n be the least common multiple of the odd numbers $1, 3, \ldots, 2n - 1$, and define the *slack*

$$\sigma(n) = L_n\Big(1 - \sum_{k=1}^{n-1} \frac{1}{2k+1}\Big) = L_n - \sum_{k=1}^{n-1} \frac{L_n}{2k+1},$$

an integer with the sign of $1 - (\frac{1}{3} + \frac{1}{5} + \cdots + \frac{1}{2n-1})$. A revolution is four quarter-turns; two revolutions are eight.

Proposition 3.9 (Once around holds, twice around slips)**.** $\sigma(4) = 34$ *and* $\sigma(8) = -982$*. The factor* $1 - \sum_{k=1}^{n-1} 1/(2k+1)$ *is strictly decreasing in* n*; it is* $2021/45045 > 0$ *at* $n = 7$ *and* $-982/45045 < 0$ *at* $n = 8$*, so* $\sigma(n) > 0$ *for* $1 \leq n \leq 7$ *and* $\sigma(n) < 0$ *for every* $n \geq 8$*.*

Proved here.

Proof. $L_4 = \operatorname{lcm}(1, 3, 5, 7) = 105$ and $105 - 35 - 21 - 15 = 34$. $L_8 = \operatorname{lcm}(1, 3, \ldots, 15) = 45045$ and $45045 - (15015 + 9009 + 6435 + 5005 + 4095 + 3465 + 3003) = 45045 - 46027 = -982$. Each step

in n subtracts a positive term from the factor. $L_7 = 45045$ as well, and $45045 - (15015 + 9009 + 6435 + 5005 + 4095 + 3465) = 2021$. $\square$

These are the signs that govern the Borwein integrals: for $n \le 7$,

$$\int_0^\infty \prod_{k=0}^{n-1} \frac{\sin(x/(2k+1))}{x/(2k+1)}\,\mathrm{d}x = \frac{\pi}{2},$$

the pattern holding exactly while $\frac{1}{3} + \cdots + \frac{1}{2n-1} < 1$, and at $n = 8$ the sum first exceeds one and the integral falls short of $\pi/2$ by about 2.3×10^{-11}. *Standard* [3]. The counts four and eight are the quarter-turns of one and two revolutions, and nothing else: no length of any tape enters this proposition. The slack changes sign once and never changes back.

The spinor is periodic. Three things can be meant by "the state returns," and they are separated before any of them is counted.

Proposition 3.10 (Once around flips, twice around returns)**.** *For $\theta \in \mathbb{R}$ let*

$$U(\theta) = \exp\!\left(-\tfrac{i\theta}{2}\sigma_3\right) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix},$$

the action on two-component spinors of the rotation by θ about the third axis.

(a) $U(2\pi) = -I$ and $U(4\pi) = I$; $U(\theta) = I$ if and only if $\theta \in 4\pi\mathbb{Z}$, and $U(\theta) = \pm I$ if and only if $\theta \in 2\pi\mathbb{Z}$.

(b) $U(\theta)\psi = (e^{-i\theta/2}\psi_1,\, e^{i\theta/2}\psi_2)$: the two components acquire opposite phases. If both components of ψ are nonzero, $U(\theta)\psi$ is a multiple of ψ if and only if $\theta \in 2\pi\mathbb{Z}$; if one component vanishes, ψ is an eigencomponent *and $U(\theta)\psi$ is a multiple of ψ for every θ.*

(c) At the quarter-turns $\theta_k = k\pi/2$ the eigencomponent $\psi = (1, 0)$ acquires the phase $e^{-ik\pi/4}$, a primitive eighth root of unity raised to k; it equals 1 if and only if eight divides k.

Proved here: $e^{\mp i\pi} = -1$, $e^{\mp 2\pi i} = 1$; $e^{-i\theta/2} = e^{i\theta/2}$ if and only if $\theta \in 2\pi\mathbb{Z}$; and $e^{-i\pi/4}$ has order eight. So the vector ψ returns at 4π; the ray $[\psi]$ of a state with both components nonzero returns at 2π; the ray of an eigencomponent never leaves. The sign at 2π is a common phase of ψ, invisible to every observable of ψ alone. It is visible as a relative phase, when ψ is compared coherently with an unrotated copy of itself, and only then (Proposition 3.11). The 4π return has been observed in exactly that way, in neutrons split between two paths of which one is carried around by a magnetic field. *Standard* [21, 24].

Proposition 3.11 (An observable that rotates)**.** *For every θ,*

$$U(\theta)\sigma_1 U(\theta)^* = \cos\theta\,\sigma_1 + \sin\theta\,\sigma_2, \qquad U(\theta)\sigma_3 U(\theta)^* = \sigma_3.$$

Hence the off-diagonal observable σ_1 returns exactly when $\theta \in 2\pi\mathbb{Z}$, at the quarter-turns exactly when four divides k, and $\frac{1}{2}\operatorname{tr}(\sigma_1 U(\theta)\sigma_1 U(\theta)^*) = \cos\theta$; the diagonal observable σ_3 returns at every angle. The vector observation $O(\psi) = \big(\langle\psi, \sigma_j\psi\rangle\big)_{j=1,2,3}$ satisfies

$$O \circ U(\theta) = R_3(\theta) \circ O, \qquad O(-\psi) = O(\psi),$$

where $R_3(\theta)$ is the rotation of $\mathbb{R}^3$ by θ about the third axis; $U(\theta)$ and $-U(\theta)$ induce the same $R_3(\theta)$. For every unit spinor ψ, $\operatorname{Re}\langle\psi, U(\theta)\psi\rangle = \cos(\theta/2)$.

Proved here.

Proof. The $(1,2)$ entry of $U\sigma_1U^*$ is $e^{-i\theta/2}e^{-i\theta/2} = e^{-i\theta} = \cos\theta - i\sin\theta$, the $(1,2)$ entry of $\cos\theta\,\sigma_1 + \sin\theta\,\sigma_2$; the $(2,1)$ entries are the conjugates, and the diagonals vanish. σ_3 is diagonal and commutes with U. The trace follows from $\operatorname{tr}\sigma_1^2 = 2$ and $\operatorname{tr}\sigma_1\sigma_2 = 0$. For the observation, $O(U\psi)_j = \langle\psi, U^*\sigma_j U\psi\rangle$ and $U(\theta)^*\sigma_1U(\theta) = \cos\theta\,\sigma_1 - \sin\theta\,\sigma_2$, $U(\theta)^*\sigma_2U(\theta) = \sin\theta\,\sigma_1 + \cos\theta\,\sigma_2$, which is the action of $R_3(\theta)$ on the first two coordinates. Last, $\langle\psi, U\psi\rangle = |\psi_1|^2e^{-i\theta/2} + |\psi_2|^2e^{i\theta/2}$, whose real part is $\cos(\theta/2)$. □

So the square of Definition 2.5 closes. The vector reading is represented by a rotation; nothing fails to commute; and what the representation loses is the sign, the two-point fibre of $U \mapsto R_3$. The coin is that fibre's receipt. A diagonal observable is a control that cannot show the pattern, since it returns at every count; σ_1 shows it. The relative phase $\cos(\theta/2)$, with period 4π, is the quantity the neutron interferometer reads.

Proposition 3.12 (Clock and clock complement)**.** *For $k \geq 1$ let $H(k)$ be the statement "sameness holds at k quarter-turns," that is $\sigma(k) > 0$, and $B(k)$ the statement "the spinor has returned at k quarter-turns," that is $U(\theta_k) = I$. At $k = 4$, H is true and B is false; at $k = 8$, H is false and B is true. Exactly one of the two holds at every k from 1 to 8. At $k = 9, \ldots, 15$ neither holds; for $k \geq 9$ exactly one holds if and only if eight divides k.*

Proved here, from Propositions 3.9 and 3.10: $H(k)$ holds exactly for $k \leq 7$ and $B(k)$ exactly when eight divides k. The result is the comparison at the two counts: where one holds the other slips, at four and at eight, and through the first two revolutions. It is not a common evolution. The slack changes sign once and the spinor is periodic, and from the ninth quarter-turn on the two readings agree in failing. *Proposed*: the slack is the tally's reading once around and twice around; the sign of the state is the register's sign; and the eight at which the Borwein pattern breaks and the eight at which the spinor returns are one eight. A common clock would be a map from quarter-turns to slack counts preserving both readings, or a dynamics of which both are readings; neither is constructed here.

Proposition 3.13 (One exponential, two readings)**.** *The positional reading of n identical unit cells is $2^n - 1$ (Proposition 1.5), and*

$$2^n - 1 = e^{n\log 2} - 1.$$

The cell read never changes; the count of readings does.

Proved here; it is the numerical card's Remark E.1. *Proposed*, there and here: indexed by time the exponential is the late-time growth of a global slicing; indexed by depth it is the growth of a warp factor in a radial coordinate; which it is depends only on how the index is read.

4 The bent card: receipt, slip, floor

In the game the bent card is the one with a corner turned up, so that the mark believes the quarter can be followed. The bend is a receipt supplied from outside, not a reading of the card.

The receipt is Definition 2.4. The electron's receipts, as the cards carry them: the sheet sign (which of two patches a point of the ball lifts to), the antipodal bit (which of z and $-z$ gave a

paired reading), the branch (which of two roots a residual selects), and the sign after one revolution (Proposition 3.10). Each is one bit where the fibre has two points; the geometric card computes the conditional entropies of its finite readers as 2/3 bit and one bit.

Proposition 4.1 (Four twofolds)**.** *The following have two elements each and are different objects.*

(i) The fibre $\{z, -z\}$ *of a paired reading: two states, one output.*

(ii) The kernel $\{I, -I\}$ *of the double cover* $\mathrm{SU}(2) \to \mathrm{SO}(3)$*,* $U \mapsto R_U$*: two descriptions of one rotation. On unit quaternions,* $(a, b) \mapsto (q \mapsto aqb^{-1})$ *is a homomorphism from pairs of unit quaternions onto* $\mathrm{SO}(4)$ *acting on* $\mathbb{R}^4$*, with kernel* $\{(1,1), (-1,-1)\}$*; its restriction to the diagonal* $a = b$ *fixes the real axis and acts on the imaginary quaternions as* $\mathrm{SO}(3)$*, with kernel* $\{1, -1\}$*. Spatial rotation is the diagonal.*

(iii) The two sectors of the finite clock of Proposition 4.2: two orbits of a group action.

(iv) The sign under exchange of two identical fermions, $\Psi(x_2, x_1) = -\Psi(x_1, x_2)$*: a rule on two-particle states.*

They occur in different spaces and act on different things. Agreement in the number of retained bits does not identify their contents, and a two-element fibre alone imposes no antisymmetry; exclusion requires the exchange rule (iv) on multiparticle states.

Proved in the geometric card for (i) and (iii) (its Section 3.4 and Propositions 3.3 and 3.4); *standard* for (ii) [11, 26] and (iv) [23]. In the game, the mark's twofold is which card; the dealer's twofold is two names for one throw. The press check of Section 9 inverts different maps and never confuses them. The Hopf account is Lyons's [14].

Proposition 4.2 (The index-two lattice and the two sectors)**.** *On the Clifford torus the two selected great-circle families have winding vectors* $(1, 1)$ *and* $(1, -1)$*, whose determinant is* -2*: they span an index-two sublattice, and a pair of windings lies in it exactly when its coordinates have the same parity. In the six-position clock the two diagonal steps generate a group of eighteen elements with two orbits on the thirty-six states, distinguished by* $j + k$ *modulo two. An antipodal pair* (j, k) *and* $(j + 3, k + 3)$ *lies in one sector and is separated by one retained bit.*

Proved in the geometric card (its Propositions 3.3 and 3.4 and Section 6.3). *Proposed*: the sector is up or down, and the bit is the coin. Two states over one output in one sector, told apart by one bit, is the shape the exclusion principle has in the finite clock; the identification with the physical principle needs the exchange rule of Proposition 4.1(iv), and is not derived.

Proposition 4.3 (Six and eight)**.** *The positions of the six-position clock are sixth roots of unity. The quarter-turn samples* $e^{-ik\pi/4}$ *of the eigencomponent's phase are eighth roots of unity. The least group of roots of unity containing both is the group of* 24*th roots, since* $\mathrm{lcm}(6, 8) = 24$*.*

Proved here. The clock and the quarter-turn sampling are two samplings of one round phase at different steps, $2\pi/6$ of the clock's angle and $\pi/4$ of the spinor's half-angle. A common account of their parameters would sample at the common refinement; neither companion card does, and the phase of Definition 7.1 is taken round.

Proposition 4.4 (Affixing a quarter to the bent card)**.** *Fill the slot of a carrier with any record. By Definition 1.2 no decision queries the slot, so every reader in the length family is unchanged, and so is every mark placed on the carrier (Proposition 1.8). Only the click reader can tell, and it reads the affixed fact, not where the fact came from: the tally and the click count agree exactly when every affixed fact holds (Proposition 1.5).*

Proved here. Measured, from the Quarter's report of the middle card: a carrier whose index is itself a carrier, dealt through every step, is identical to the plain carrier for the first twelve steps at no cost, and differs from the step at which one construction lifts its quantum one universe above the box, and from every later step. *Proposed*: the second counting layer is the first with a quarter affixed at every step. What is fixed under the card is supplied, not read, which is Theorem 2.7 at the table.

4.1 The slip and the floor

A metered answer has the shape of a bracket, a lower bound and a slip bound; a point is the shape only logic has. A meter has a floor, and below the floor it reads itself. So a metered turn stops one arm short of the point, on this side of the slip, and that is not a failure to converge; it is the shape the answer has. *Measured*: the middle card's disagreeing arm names two distinct bounds, a bracket and not a point, and its one dimensionless measurement is heartbeats per cell, a Courant number, which differs between machines and so is a measurement rather than a reading. The floor that does not differ between machines is the uncertainty principle: for any function and its Fourier transform the product of the widths is at least one half. *Standard* [12]. *Proposed*: the record can bracket the electron and never point at it. The point is where a number would be, and the card does not go there. This is the reason at its third level, the game's own: the quarter is glued to the table, and the throw never moves it.

4.2 The loop

The record of a term that never stops is a one-symbol tape; the loop writes no quarter, since a record of one closed term repeated determines nothing about any variable; and the only harm a loop can do is to be read as a value, which a record system that accepts only normal forms never does. *Proved in the bent card* (its Propositions 3.2 to 3.4), on Dijkstra's objection [8] and Barendregt's calculus [2]. In the game, the dealer's hands never stop. The mark who reads the motion as a value is the mark.

5 The accomplice and the force: the record

The middle card ends with two acts. A reviewer makes demands from the top page down to the first, pairing at each page a claim found by search against the same claim handed over by name: this is the accomplice, and the descent is a boundary-value problem, elliptic in character, with the datum above each stage already in hand, and it terminates. A publication is built from the first page up to the top and read once there: this is the force, and the ascent is an initial-value problem, hyperbolic in character, with no signal from the last page reaching a reader placed before it. *Recorded*, from the numerical card's Remark E.2; the reading of the two directions as the two holographies is *proposed* there [15, 25, 22] and is not used here.

Definition 5.1 (This card's record)**.** The *register* is the coin: the retained sign after one revolution, Proposition 3.10, carried unchanged through every entry. The *counter* is the number k of quarter-turns, from 0 to 8, increased by exactly one per entry and never skipped. The *trace* is the sequence of nine entries; entry k quotes the pair

$$A_k:\ U(\theta_k)\sigma_1 U(\theta_k)^* = \sigma_1, \qquad B_k:\ U(\theta_k) = I,$$

the first saying that the rotating vector observable has returned, the second that the spinor has, and writes the proposition $A_k \leftrightarrow B_k$ without deciding it. The *verdict* is read once, at the top, Section 9: for two readings X, Y of the register, *slipped* is the decidable proposition $\neg(X \leftrightarrow Y)$, decided from the two readings' own decisions and never by an appeal to choice; the complete verdict is the triple (slipped, value, velocity).

Remark (Velocity is the penalty parameter). The counter at the top is the cost of carrying the sign constraint exactly: one step per quarter-turn, eight to return. A penalty would relax the constraint at a finite wave speed; the record does not relax it, and pays in entries instead. *Proposed*; the definition of the counter is *measured* from the middle card, whose record lifts its register one universe per rung and reports the top universe as its velocity.

Remark (Inhabitation is not acceptance)*.* That the record exists, nine entries each with its pair, carrying one register through a counter that reaches eight, is well-formedness. It says nothing about the verdict: the same form carries a record whose two readings disagree at the top. Acceptance is a property of the trace, read once. *Recorded*, from both surrounding cards' Section 8.3.

6 Transport: from a phase to the commutator

The carrier carries. Now it moves. Compare the phase here with the phase there, and you need a rule. Make the rule covariant, and it is a connection. Go around and come back, and the connection keeps a record of the trip. Go around two ways, and the difference is the commutator on the cover of this card.

Setting. Spacetime is Minkowski space $M = \mathbb{R}^{1,3}$ with $\eta = \mathrm{diag}(1,-1,-1,-1)$, in natural rationalized units $\hbar = c = 1$, as on the numerical card's Section 2.4. *Assumption*: a Lorentzian background. The device supplies none; it is *standard*.

Definition 6.1 (Phase-carrying state)**.** A *phase-carrying state* is a smooth field $\psi : M \to \mathbb{C}^N$ together with the action of the phase group at every point,

$$\psi(x) \longmapsto e^{-ie\chi(x)}\psi(x),$$

for smooth real χ and a fixed real parameter e, the *coupling*. The transformation is *global* when χ is constant and *local* otherwise. The *density observation* at x is $O_x(\psi) = \psi(x)^\dagger\psi(x)$.

The fibre of O_x contains the whole phase orbit of $\psi(x)$: the phase at a point is a coordinate on a fibre, carried by the state and not queried by the density. It is the slot of Definition 1.2, read round. *Assumption*: a continuous phase group acting on a complex field. The device supplies a two-valued sign receipt and a six-position clock (Propositions 3.10 and 4.2), that is, a finite subgroup of the phases; the circle is *standard* and not supplied.

Proposition 6.2 (Comparison needs a rule)**.** *Under a local phase change,* $\psi(y)^\dagger\psi(x) \mapsto e^{ie(\chi(y)-\chi(x))}\psi(y)^\dagger\psi(x)$*. It is invariant under every global change, and under a local change only where* $\chi(x) = \chi(y)$*.*

Proved here, by substitution. So the naive comparison of two points is a level-(a) datum of Definition 2.8: it names a representative of the phase at each point. To compare without lifting, a rule must carry the fibre at x to the fibre at y.

Definition 6.3 (Transport; connection)**.** A *connection* is a smooth real one-form $A = A_\mu\,\mathrm{d}x^\mu$ on M. Its *transport* along a path γ from x to y is

$$W_\gamma = \exp\Big(-ie\int_\gamma A_\mu\,\mathrm{d}x^\mu\Big),$$

a unit complex number carrying the fibre at x to the fibre at y. Under a local phase change the connection transforms by $A_\mu \mapsto A_\mu + \partial_\mu\chi$.

Assumption: that comparison is carried by a connection. The device supplies no transport between points of a manifold. *Measured*: the middle card counts the closed walks on the graph of its arms, shows that every closed walk crosses an even number of times, and declares half a turn per crossing, forward; under that declaration every closed walk turns a whole number of times. A closed walk returns to its vertex; a holonomy returns a phase. That the middle card's count of turns is the holonomy of a connection is *proposed*, and the map from crossings to phases is the one that would have to be constructed.

Definition 6.4 (State, gauge orbit, gauge-invariant observable)**.** From here on the *state* is the pair (A, ψ). A phase change χ acts on it by $(A, \psi) \mapsto (A + \mathrm{d}\chi,\, e^{-ie\chi}\psi)$. The *gauge orbit* of (A, ψ) is the set of its images under all phase changes, and the *orbit map* π sends a pair to its orbit. A *gauge-invariant observable* is a function of the pair that is constant on gauge orbits, equivalently a function on the quotient: a level-(b) datum of π in the sense of Definition 2.8.

Proposition 6.5 (Transport is covariant)**.** *Under* $\psi \mapsto e^{-ie\chi}\psi$*,* $A \mapsto A + \mathrm{d}\chi$*:*

(a) $W_\gamma \mapsto e^{-ie\chi(y)}\,W_\gamma\,e^{ie\chi(x)}$*, and the transported comparison* $\psi(y)^\dagger W_\gamma\psi(x)$ *is invariant;*

(b) the infinitesimal transport defines the covariant derivative

$$D_\mu\psi(x) = \lim_{\epsilon\to 0}\frac{W_{x\leftarrow x+\epsilon e_\mu}\,\psi(x+\epsilon e_\mu) - \psi(x)}{\epsilon} = (\partial_\mu + ieA_\mu)\psi(x),$$

where $W_{x\leftarrow x+\epsilon e_\mu}$ *is the transport along the segment from* $x + \epsilon e_\mu$ *to* x*;*

(c) D_μ *is covariant: with primes for the transformed fields,* $D'_\mu\psi' = e^{-ie\chi}D_\mu\psi$*.*

Proved here.

Proof. (a) $\int_\gamma \partial_\mu\chi\,\mathrm{d}x^\mu = \chi(y) - \chi(x)$, so $W'_\gamma = W_\gamma e^{-ie(\chi(y)-\chi(x))}$; then

$$\psi'(y)^\dagger W'_\gamma\psi'(x) = e^{ie\chi(y)}\psi(y)^\dagger e^{-ie\chi(y)}W_\gamma e^{ie\chi(x)}e^{-ie\chi(x)}\psi(x).$$

(b) $W_{x\leftarrow x+\epsilon e_\mu} = \exp(ie\epsilon A_\mu(x)) + O(\epsilon^2) = 1 + ie\epsilon A_\mu(x) + O(\epsilon^2)$; expand $\psi(x+\epsilon e_\mu)$ to first order. (c)

$$\begin{aligned}(\partial_\mu + ieA_\mu + ie\partial_\mu\chi)(e^{-ie\chi}\psi) &= e^{-ie\chi}(\partial_\mu\psi - ie\partial_\mu\chi\,\psi + ieA_\mu\psi + ie\partial_\mu\chi\,\psi)\\ &= e^{-ie\chi}D_\mu\psi.\end{aligned}$$

□

Covariance is a commuting square (Definition 2.5): multiply by the phase and then differentiate with the transformed connection, or differentiate and then multiply, and the two routes agree. The square closes exactly, and no phase representative was named on either route.

Proposition 6.6 (Curvature)**.** *Define $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. Then F is invariant under $A \mapsto A + \mathrm{d}\chi$. For a closed path γ the transport W_γ is invariant, and if γ bounds a surface Σ then $W_\gamma = \exp(-ie\int_\Sigma \frac{1}{2}F_{\mu\nu}\,\mathrm{d}x^\mu \wedge \mathrm{d}x^\nu)$; around the rectangle with sides ϵe_μ then δe_ν, traversed in that order, $W = 1 - ie\,\epsilon\delta F_{\mu\nu} + O(\epsilon\delta(\epsilon+\delta))$. On a simply connected region, W_γ depends only on the endpoints of γ for every γ if and only if $F = 0$ there.*

Proved here, from Stokes's theorem; *standard* [23].

Proof. $\partial_\mu\partial_\nu\chi$ is symmetric. For closed γ the factors $e^{\mp ie\chi(x)}$ of Proposition 6.5(a) cancel. Stokes gives $\oint_\gamma A = \int_\Sigma \mathrm{d}A$ and $\mathrm{d}A = \frac{1}{2}F_{\mu\nu}\,\mathrm{d}x^\mu \wedge \mathrm{d}x^\nu$; the rectangle is the case of small Σ. Path independence for all paths is $\oint A \in (2\pi/e)\mathbb{Z}$ for every closed path; shrinking the loop forces $\oint A = 0$, which on a simply connected region holds for all loops exactly when $\mathrm{d}A = 0$. □

The curvature is the obstruction to path-independent transport: the record of a closed trip, kept by the connection and read by no phase representative.

Theorem 6.7 (The commutator of representation)**.** *For every smooth ψ and connection A,*

$$[D_\mu, D_\nu]\psi = ie\,F_{\mu\nu}\,\psi.$$

Proved here.

Proof. $D_\mu D_\nu\psi = \partial_\mu\partial_\nu\psi + ie(\partial_\mu A_\nu)\psi + ieA_\nu\partial_\mu\psi + ieA_\mu\partial_\nu\psi - e^2A_\mu A_\nu\psi$. Exchanging μ and ν and subtracting removes every term symmetric in $\mu\nu$ and leaves $ie(\partial_\mu A_\nu - \partial_\nu A_\mu)\psi$. □

This is the commutator the title promised. Transport along μ then ν, and along ν then μ, are two transformations of the state; they fail to commute, and the failure is multiplication by the curvature. It is covariant, since multiplication by F commutes with the phase. It is not a sign lost in a fibre, as the spinor's is (Proposition 3.11); it is a field on spacetime, and it can be observed, as a phase difference between two paths (Proposition 8.1).

6.1 The action

Covariance says how the pieces must transform. It does not say which pieces. The following enter here, and each is an assumption. *Assumption*: the spinor representation: ψ is a Dirac spinor, $N = 4$, with fixed gamma matrices satisfying $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}I$ and $(\gamma^\mu)^\dagger = \gamma^0\gamma^\mu\gamma^0$, conjugate $\bar\psi = \psi^\dagger\gamma^0$, and $\psi, \bar\psi$ independent anticommuting variables in the variation. The device supplies the sign after one revolution, which a spin-$\frac{1}{2}$ representation has and which does not determine one. *Assumption*: the kinetic terms of Maxwell and Dirac. *Assumption*: a mass parameter m. The coupling e is the parameter of Definition 6.1.

Definition 6.8 (The Maxwell–Dirac action)**.**

$$S[A,\psi,\bar\psi] = \int_M \Big[-\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar\psi(i\gamma^\mu D_\mu - m)\psi\Big]\,\mathrm{d}^4x = \int_M \Big[-\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar\psi(i\gamma^\mu\partial_\mu - m)\psi - e\bar\psi\gamma^\mu A_\mu\psi\Big]\,\mathrm{d}^4x.$$

The second form is the first with $D_\mu = \partial_\mu + ieA_\mu$ expanded: $\bar\psi i\gamma^\mu(ieA_\mu)\psi = -e\bar\psi\gamma^\mu A_\mu\psi$. It is the numerical card's bare action, letter for letter. The coupling enters only through the connection: minimal coupling is the covariant derivative of Proposition 6.5 written into the Dirac term.

Proposition 6.9 (The stationary system)**.** *Varying S in the independent fields $A, \psi, \bar\psi$ with compactly supported variations gives*

$$(i\gamma^\mu D_\mu - m)\psi = 0, \qquad i(\partial_\mu\bar\psi - ieA_\mu\bar\psi)\gamma^\mu + m\bar\psi = 0, \qquad \partial_\mu F^{\mu\nu} = e\bar\psi\gamma^\nu\psi,$$

and $\partial_{[\lambda}F_{\mu\nu]} = 0$ holds because $F = \mathrm{d}A$. On solutions the current $j^\mu = \bar\psi\gamma^\mu\psi$ is conserved, $\partial_\mu j^\mu = 0$; the $\nu = 0$ Maxwell equation is Gauss's constraint $\partial_i F^{i0} = e\psi^\dagger\psi$. The action is invariant under the global phase change, and j^μ is its Noether current; for fields decaying at spatial infinity the charge $Q = \int \psi^\dagger\psi\,\mathrm{d}^3x$ is constant in time. Every term of S has at most three independent-field factors, so $D^4 S = 0$.

Proved in the numerical card (its Section 2.4, where each equation is derived and the cubic count is made coefficient by coefficient in the Grassmann algebra); *standard* for the Noether identification [18, 23]. This is item (4) of Proposition 1.11, supplied: a continuous group, an action invariant under it, and a current conserved on the stationary system. The charge is Q, and eQ is the source in Gauss's law.

Proposition 6.10 (Covariance does not select the action)**.** *The Pauli term $\bar\psi\sigma^{\mu\nu}\psi F_{\mu\nu}$, with $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$, is invariant under local phase changes, is a Lorentz scalar, and has three independent-field factors. The term $\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}$ is invariant and quadratic, and is a total derivative. Every real m is admissible. So local phase covariance, Lorentz covariance and degree at most three together admit actions other than Definition 6.8.*

Proved here: $\bar\psi \mapsto \bar\psi e^{ie\chi}$ and $\psi \mapsto e^{-ie\chi}\psi$ cancel in every bilinear, and F is invariant (Proposition 6.6); the count of factors is read off. The additional criterion that excludes the Pauli term in standard practice is power counting: its coefficient has the dimension of a length, which is not admissible in a renormalizable action. *Standard* [19]. The device supplies no such criterion; its selection of the minimal action is *assumed*.

Proposition 6.11 (The field propagates)**.** *With $\psi = 0$ the stationary system is $\partial_\mu F^{\mu\nu} = 0$; in a Lorenz gauge $\partial_\mu A^\mu = 0$ it is $\Box A^\nu = 0$. A plane wave $A^\nu = \varepsilon^\nu e^{-ik\cdot x}$ solves it if and only if $k\cdot k = 0$, so that $\omega = |\mathbf{k}|$ and the wave moves at the speed of light; the gauge condition is $k\cdot\varepsilon = 0$; and $\varepsilon \mapsto \varepsilon + c\,k$ changes A by a gradient and leaves F unchanged. For $k \neq 0$ two independent polarizations remain.*

Proved here, by substitution: $\Box e^{-ik\cdot x} = -k\cdot k\, e^{-ik\cdot x}$; $\partial_\mu A^\mu = -ik\cdot\varepsilon\, e^{-ik\cdot x}$; $k_\mu k_\nu - k_\nu k_\mu = 0$; and the solutions of $k\cdot\varepsilon = 0$ form a three-dimensional space containing k. *Standard* [23]. In the classical theory a current is the source of the field, $\partial_\mu F^{\mu\nu} = ej^\nu$, and the interaction term $-ej^\mu A_\mu$ is where the two exchange energy. In the quantum theory a photon is one quantum of a normal mode of this free field, and emission and absorption are the cubic vertex $-e\bar\psi\gamma^\mu A_\mu\psi$ at first order in e; *standard* [19, 23], and it requires the quantization named below.

What the device supplies, and what is assumed. The chain is complete from the phase to the stationary system.

Ingredient	Supplied by the device, or assumed	Calibration
Phase-carrying state	Device: a slot carried and not queried, a sign receipt, a six-position clock. Assumed: a circle of phases on a complex field.	*Proposed* that the slot is the phase; *standard* for the circle
Comparison rule	Assumed; that one is needed is Proposition 6.2.	*Proved here*
Transport, connection	Assumed. Device: closed walks with an even count of crossings and a declared half-turn per crossing.	*Measured* for the count; *proposed* that it is a holonomy
Curvature, commutator	Derived from the connection.	*Proved here*
Lorentzian background	Assumed.	*Standard*
Spinor representation	Assumed; the device's sign after one revolution is consistent with it and does not determine it.	*Standard*
Kinetic terms, minimal coupling	Assumed; not selected by covariance and degree (Proposition 6.10).	*Standard*
Mass m	Assumed parameter.	—
Coupling e	Device: the numerical card's certified reading, Definition 7.1(viii).	*Recorded*
Stationary system	Derived from the action.	*Proved in the numerical card*
Quantization, regularization, renormalization	Not supplied; required for quantum electrodynamics proper.	*Standard*

The result of this section is therefore a reconstruction of classical Maxwell–Dirac theory from a phase-carrying carrier, with its assumptions on the table. Quantum electrodynamics proper is the quantization of this action with anticommuting fermion fields, together with a regulator and a renormalization prescription that fixes m and e from measurements [19]; the numerical card's Section 2.4 marks the same border, and none of it is supplied here.

The transported comparison $\psi(y)^\dagger W_\gamma \psi(x)$, the holonomy of a closed path, the curvature F, the current j^μ and the charge Q are gauge-invariant observables (Definition 6.4): level-(b) data of the orbit map π, by Propositions 6.5 and 6.6 and because $\bar\psi \mapsto \bar\psi e^{ie\chi}$ cancels $\psi \mapsto e^{-ie\chi}\psi$ in every bilinear. The chain never names a phase representative. The density O_x is one more observable on the same quotient, and a coarser one: the pairs $(0, \psi)$ and (A, ψ) with $\mathrm{d}A \neq 0$ have the same density at every point and different curvature, and equal densities do not fix the current (the example after Definition 7.1). Gauge invariance is factoring through π, not through the density. A choice of gauge, such as the Lorenz condition, is a choice of a point on each orbit and is used only to display solutions. *Proved here.*

7 The electron, dealt

A carrier, a phase, a transport, a curvature, an action and its stationary system. Deal it.

Definition 7.1 (The electron of this card)**.** The electron of this card is the following data.

(i) A *carrier* (Definition 1.2) whose slot is the phase coordinate of (vi): carried, and queried by no decision.

(ii) A *state*: a Dirac spinor field ψ on M with conjugate $\bar{\psi}$, and a connection A (Section 6).

(iii) A *transformation law*: under a phase change, $\psi \mapsto e^{-ie\chi}\psi$ and $A \mapsto A + \mathrm{d}\chi$; under the rotation by θ about the third axis, $\psi \mapsto S(\theta)\psi$ with $S(\theta) = \mathrm{diag}(U(\theta), U(\theta))$ in the chiral representation, U as in Proposition 3.10.

(iv) An *observation map*: $O(\psi) = \big(\psi^\dagger\psi,\ \psi^\dagger\Sigma_j\psi\big)$, the charge density and the spin density, with $\Sigma_j = \mathrm{diag}(\sigma_j, \sigma_j)$.

(v) A *coin*: the receipt that resolves the twofold lift $\{\psi, -\psi\}$, which lies in one fibre of O, read as the sign of a coherent comparison of ψ with a copy of itself (Proposition 3.11); in the finite realization, the antipodal bit of Proposition 4.2.

(vi) A *phase*: a point of the circle; in the finite realization, a point of the six-position clock.

(vii) A *rule*: the stationary system of the Maxwell–Dirac action, Proposition 6.9.

(viii) A *coupling*: $e > 0$ with

$$\frac{4\pi}{e^2} = 137.011\,290\,5\ldots,$$

the numerical card's certified reading.

(ix) A *count* of quarter-turns, k, with $\theta_k = k\pi/2$.

This is a definition, and it makes no physical claim. The value in (viii) is quoted: *recorded* from the numerical card's worked reading (its Chapter 6), where it is certified by a bisection bracket and proved there; this card does not compute it, and it does not lift it. In the rationalized natural units of Section 6 the dimensionless coupling is $e^2/4\pi = 1/137.011\,290\,5\ldots$, so that $e^2 \approx 0.0917178$ and $e \approx 0.302849$. The four fields a margin of the middle card lists for the electron, a calibration of minus one, a phase, two energies whose difference is the action, and a fact that is up or down, are (v), (vi), (vii) and the sector of the coin; *measured*.

The pair $\{\psi, -\psi\}$ is not the whole fibre of O. Take the chiral representation $\gamma^0 = \left(\begin{smallmatrix} 0 & I \\ I & 0 \end{smallmatrix}\right)$, $\gamma^i = \left(\begin{smallmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{smallmatrix}\right)$, in which $\Sigma_j = \frac{i}{2}\epsilon_{jkl}\gamma^k\gamma^l$ and $S(\theta)$ is the block matrix of (iii). The states $\psi = (1,0,0,0)$ and $\varphi = (0,0,1,0)$ both have $O = (1;\, 0,0,1)$, and φ is no phase multiple of ψ; their currents differ, $\bar{\psi}\gamma^3\psi = -1$ and $\bar{\varphi}\gamma^3\varphi = +1$. *Proved here*, by evaluation. The coin resolves the specified twofold lift, and not every ambiguity of the fibre.

Proposition 7.2 (What the definition proves)**.** *From Definition 7.1 and the cited propositions:*

(a) Rotation. *$S(2\pi) = -I$ and $S(4\pi) = I$; $O(S(\theta)\psi)$ is $O(\psi)$ with its spin density rotated by $R_3(\theta)$, and $O(-\psi) = O(\psi)$. The observation returns after one revolution, the state after two; and* $\mathrm{Re}\langle\psi, S(\theta)\psi\rangle = \cos(\theta/2)\,\psi^\dagger\psi$ *(Propositions 3.10 and 3.11, block by block).*

(b) Phase. *O is invariant under every phase change; $D_\mu\psi$ is covariant; $[D_\mu, D_\nu]\psi = ieF_{\mu\nu}\psi$ (Proposition 6.5, Theorem 6.7).*

(c) Dynamics. *On the rule: the two Dirac equations, Maxwell's equations with source ej^ν, conservation of j^μ, Gauss's constraint, and a conserved charge Q (Proposition 6.9); the free field propagates at the speed of light in two polarizations (Proposition 6.11).*

(d) Charge and count. *The charge of (c) is the Noether charge of the global phase. The count of cells of Proposition 1.10 is invariant under erasure and is not that charge; that it reads it is* proposed *(Proposition 1.11).*

(e) Two over one. *The states ψ and $-\psi$ have one observation and are separated by one bit, the coin. Exclusion is not a consequence: it requires the exchange rule of Proposition 4.1(iv), which is* assumed *with the anticommuting fields.*

(f) Scalar reading. *If a scalar reading of the rule is a quadratic residual with two distinct real roots, which is a hypothesis and not a consequence of degree at most three, then Propositions 3.1–3.4 apply to it: its first derivative at the reading squares to the discriminant* (3.2)*, and Newton's error is the acceleration term* (3.3)*.*

(g) Blindness of the mark. *The mark that carries any of these facts fits an empty carrier exactly as well (Proposition 1.8).*

Proved here, as consequences of the definition and the cited propositions, with (c) *proved in the numerical card*. Each holds for every value of e; none uses the digits of (viii). Each physical reading of a clause is *proposed*. Electrodynamics as a derivative is the numerical card's identification in its section "The derivative of the modified field equations": for a background ψ_0 and a differentiable map Φ from the electrodynamic field space to the gravitational one,

$$DR_{\mathrm{QED}}(\psi_0) = D\Phi(\psi_0)^* \circ DR_{\mathrm{GR}}\big(\Phi(\psi_0)\big) \circ D\Phi(\psi_0),$$

a comparison of linearized operators in which residual values are compared by the pullback $D\Phi(\psi_0)^*$. Neither ψ_0 nor Φ is exhibited, and the identity is calibrated *proposed* there and here. The reading of (f) through it, the first-order coupling being the coefficient of the one-vertex term and the linear response, with the second variation being the one sentence the strong interaction gets, is *proposed*; a proof ends where the first derivative has been applied once more and the test clicks.

Proposition 7.3 (The record cannot tell which electron)**.** *Three carriers may be placed on the table: the electron that holds a bit in silicon; the electron of the Dirac equation; and the electron of Definition 7.1. By Propositions 1.8 and 1.9, any mark placed on one agrees in every field with the same mark placed on another, and any two readers of lengths agree on all three. The record carries the name of its carrier and cannot tell it from the others.*

Recorded. The second carrier is *standard*, and its structure is the one Section 6 assembles: the gauge action $A \mapsto A + \mathrm{d}\chi$, $\psi \mapsto e^{-ie\chi}\psi$, the two Dirac equations, the conserved current $\bar{\psi}\gamma^\mu\psi$, the Gauss constraint, and the free block whose square is $(|k|^2 + m^2)I$ [23]; the numerical card works each of these in its Section 2.4. The third carrier has the second's transformation law, observation map and rule, and the device's coupling. What supplies the electron to the record is its name, or a measurement; the record supplies neither.

Proposition 7.4 (What the electron cannot see)**.** *By Proposition 3.8, the coupling of an action is not a function of its degree hierarchy, and it is a function of its derivative. The electron of Definition 7.1 obeys a stationary system, so its coupling is in its law. Every line proved about it holds for every real e. Hence its value is seen by no proved line; it enters once, as the parameter (viii), quoted.*

Recorded. So the rule of this card, Definition 2.8, is consistent with the object: what is invariant under replacing e is proved, and what depends on its value is quoted or proposed. Three readings,

one object: the number is a representative (Corollary 2.9); the length family is blind to it (Section 1); and the degree is blind to it while the derivative carries it (this proposition). The game says the same in one sentence: the quarter is glued to the table.

The one step. The chain reaches the minimally coupled Maxwell–Dirac stationary system, with its coupling read from the device: from a phase-carrying carrier through comparison, transport, connection and curvature to the commutator, and from the commutator to the action and the stationary system the numerical card derives from it, with $e^2/4\pi = 1/137.011\,290\,5\ldots$. It stops at one step, whose scope is this: the comparison of the coupling of the assembled Maxwell–Dirac model with the laboratory's. The recommended value of the inverse fine-structure constant is $137.035\,999\,177(21)$ [17], with relative standard uncertainty 1.5×10^{-10}. The device's reading is $137.011\,290\,5\ldots$. It lies below by $0.024\,708\,7$, a relative difference of 1.80×10^{-4}, some 1.2×10^{6} standard uncertainties. At the precision the laboratory works to, about 10^{-10}, measurement excludes it. *Standard* for the recommended value; *recorded* for the difference, from the numerical card's comparison with the measured constant (its Chapter 6), where it is proved by subtraction. The identification of the coupling of Definition 7.1 with the coupling of the laboratory electron is calibrated *proposed*, and it is handed to the knower. The correspondences from the middle card to continuous transport, the slot as the phase and the count of turns as a holonomy, are not this step; each is named and calibrated where it enters (Section 6). The narrow form offered in Section 8 depends on the step.

The clock. Turn the number over. The table that weighs the steps, with weights the device counts (Proposition 10.1), is

$$M = \begin{pmatrix} 18 & 5 \\ 18 & 18 \end{pmatrix} = 18\,I + N, \qquad N = \begin{pmatrix} 0 & 5 \\ 18 & 0 \end{pmatrix}, \qquad N^2 = 90\,I.$$

The scalar part $18\,I$ is the clicks: it keeps the coin, TRUE, TRUE, TRUE. The part N changes the coin at every step and at no other time: TRUE, FALSE, TRUE, FALSE. Two ticks of N return every coin to itself, scaled by 90; it is a clock of period two. The growth rate is the sum, $\lambda_+ = 18 + \sqrt{90} = 18 + 3\sqrt{10}$, the clicks plus the clock's rate, and the reading is $\frac{324}{65}\lambda_+$:

$$137.011\,290\,5\ldots \;=\; \underbrace{89.723\,076\,9\ldots}_{18\,\cdot\,\frac{324}{65}\,:\ \mathrm{TTTT}\cdots} \;+\; \underbrace{47.288\,213\,6\ldots}_{3\sqrt{10}\,\cdot\,\frac{324}{65}\,:\ \mathrm{TFTF}\cdots}$$

Read every TRUE as TRUE and the number is $89.72\ldots$, and it is not the number. Everything above the clicks is carried by walks that alternate at every step, and by nothing else. Give each change of coin half a turn, all forward, and the clock turns once in two ticks. *Proved here*: it is the cheat sheet's Proposition 9.3, with the turns of its Proposition 9.5. The residue of the reading over the clicks is a clock. TRUE, FALSE, TRUE, FALSE. What it keeps time for is the knower's.

7.1 The photon

The field comes first. Proposition 6.11 gives it: two polarizations, the light cone, a current for a source, and, after quantization, a quantum emitted and absorbed at the cubic vertex. That is the photon of the stationary system.

Then the device's photon. The shortest tape has two ends and nothing between. The middle card's zeroed model of a single photon is a superposition of two ends, one the surface past which nothing earlier is seen and the other the singularity into which nothing later comes back, with no displacement and no velocity between them; *measured*. A photon emitted toward us from the sphere where the expansion recedes at the speed of light has, in proper distance, an approach velocity of exactly zero, and arrives only where the sphere grows; *standard* [4], recorded in the Quarter as the missing zero counted one cell late. *Proposed*: in this card the word photon names the interaction of two electrons and nothing else; a carrier emitted and absorbed by a carrier, whose tape is the two ends. The map from that tape to a normal mode of the free field of Proposition 6.11, and from the cosmological horizon to the electromagnetic construction, is not constructed here. Without it the two ends do a propagator's work by name.

8 The proposed experiment, without its number

The cheat sheet proposes an arrangement and a number for it. This card restates the arrangement and one observable, and not the number.

The observable. The phase of an interference fringe. A beam is split between two paths and recombined. On one path the spin is rotated by θ about an axis; the two paths differ in addition by a phase ϕ. For a state polarized along the rotation axis, ideal visibility and equal amplitudes, the intensity at one output is

$$I(\theta,\phi) = \frac{I_0}{2}\Big[1+\cos\big(\tfrac{\theta}{2}-\phi\big)\Big]; \tag{8.1}$$

with visibility $\mathcal{V}<1$ the cosine carries the factor $\mathcal{V}$. In the arrangement,

$$\phi = \phi_{\mathrm{AB}} + \phi_{\mathrm{dyn}} + \delta, \qquad \phi_{\mathrm{AB}} = \frac{q}{\hbar}\oint A\cdot \mathrm{d}l = \frac{q\Phi}{\hbar},$$

where ϕ_{AB} is the holonomy of Proposition 6.6 for a carrier of charge q enclosing flux Φ, with its sign fixed by the orientation of the loop [1]; ϕ_{dyn} collects the known dynamical phases of the two paths; and δ is the residue.

Proposition 8.1 (The fringe)**.** *For $\psi=(1,0)$ and output amplitude $\frac{1}{2}\big(\psi+e^{i\phi}U(\theta)\psi\big)$, the intensity is $\frac{1}{2}\big[1+\cos(\theta/2-\phi)\big]$ times the input intensity. For $\phi=0$ and every unit spinor it is $\frac{1}{2}[1+\cos(\theta/2)]$, with period 4π in θ.*

Proved here: the intensity is $\frac{1}{2}\big[1+\mathrm{Re}\big(e^{i\phi}\langle\psi,U(\theta)\psi\rangle\big)\big]$, and Proposition 3.11 gives the inner product.

Control. Neutrons, $q=0$, no residue: the fringe has period 4π in θ, the spinor's return, observed [21, 24]. *Standard.*

Setup. An electron carried twice around a carrier that is neutral and expels magnetic field. Positronium is neutral at monopole order and responds at higher multipole order; a condensed phase of it has been proposed and not yet made [20]. A superconductor expels magnetic field [16]. In general relativity a moving mass produces a field with the form of magnetism, and the frame dragging it predicts has been measured [9]. *Standard*, each. The force an orbiting charge meets from the expulsion would be a second observable; it needs an explicit model of the expelled field, which is not given, and it is not used.

Prediction, in this card's terms. The standard theory of Section 6 predicts $\delta = 0$: every phase between the paths is a holonomy or a dynamical phase. The proposal is that $\delta \neq 0$ in this arrangement and in no other, that the first revolution leaves the residue and the second takes it away or turns it over: a sign, by Proposition 3.10; a slack, by Proposition 3.9; one clock in two phases at four and at eight, by Proposition 3.12. *Proposed.*

What is owed, and where the number lives. The value of δ is the numerical card's and is a representative; it is not restated here. That card also records, from the measured anomalous moment [10, 17], that the coupling read as the fine-structure constant in general is already excluded (the one step of Section 7), and that only the narrow form, a prediction for one stated arrangement, is offered. What the arrangement still owes is a buildable carrier with the two properties, and a reason it counts.

9 The press check

The register is read once, here, by the two readings of Definition 5.1, at the top of the record and at its midpoint. Nothing beneath the head is re-read by the press check.

Reading A: the vector reading. A_k holds exactly when four divides k, by Proposition 3.11: the rotating observable σ_1 returns exactly when $\cos\theta_k = 1$, equivalently when $U(\theta_k) = \pm I$.

Reading B: the spinor reading. B_k holds exactly when eight divides k, by Proposition 3.10: $U(\theta_k) = I$ exactly then, and $\frac{1}{2}\operatorname{tr} U(\theta_k) = \cos(k\pi/4)$ is 1 exactly then.

The trace. Entry k quotes (A_k, B_k) and writes $A_k \leftrightarrow B_k$ undecided. The two numerical rows are the quantities the readings decide; the last row is the diagonal control, which returns at every count and cannot show the pattern.

k	0	1	2	3	4	5	6	7	8
$\frac{1}{2}\operatorname{tr}(\sigma_1 U \sigma_1 U^*) = \cos\theta_k$	1	0	-1	0	1	0	-1	0	1
A_k	✓				✓				✓
$\frac{1}{2}\operatorname{tr} U(\theta_k) = \cos\frac{k\pi}{4}$	1	$\frac{\sqrt{2}}{2}$	0	$-\frac{\sqrt{2}}{2}$	-1	$-\frac{\sqrt{2}}{2}$	0	$\frac{\sqrt{2}}{2}$	1
B_k	✓								✓
$U\sigma_3 U^* = \sigma_3$	✓	✓	✓	✓	✓	✓	✓	✓	✓

Proposition 9.1 (The two verdicts)**.** *At the top, $k = 8$: A_8 and B_8 both hold, so $\neg(A_8 \leftrightarrow B_8)$ is false. Slipped: false. Value: the coin, read back unchanged. Velocity:* 8*. Once around, $k = 4$: A_4 holds and B_4 fails, so $\neg(A_4 \leftrightarrow B_4)$ is true. Slipped: true. Value: the coin, now carrying the sign -1, since $U(\theta_4) = -I$. Velocity:* 4*.*

Proved here, from Propositions 3.10 and 3.11; both decisions are evaluations of a root of unity and neither uses a choice.

$$(A, B) \text{ at } 8: \ (\text{false}, \ \text{the coin}, \ 8), \qquad (A, B) \text{ at } 4: \ (\text{true}, \ -\text{the coin}, \ 4).$$

The record slips once around and does not slip twice around. The numerical card reports two triples from one register in the same way, one not slipped and one slipped by the smallest representable amount; here the smallest representable amount is a sign.

The demand-level reading. Separate from the press check and outside its rule, a second reading goes through the calibration column of this card and records, at each place, where the reading handed over by the knower says more than the reading proved. It does not decide any proposition written undecided; it lists the proposals:

- that the coupling of the carrier is the laboratory's, the one step (Section 7);
- that the carrier is the electron (Proposition 7.3);
- that the middle card's count of turns is the holonomy of a connection (Definition 6.3);
- that the two sectors of the even clock are up and down, and the bit the coin (Proposition 4.2);
- that the eight of the slack and the eight of the spinor are one eight (Proposition 3.12);
- that contents are the symmetry and cells the charge (Propositions 1.11 and 7.2(d));
- that electrodynamics is the pulled-back linearization of the modified field equations, $DR_{\mathrm{QED}}(\psi_0) = D\Phi(\psi_0)^* \circ DR_{\mathrm{GR}}(\Phi(\psi_0)) \circ D\Phi(\psi_0)$, and the second variation the strong interaction's one sentence (Proposition 7.2);
- that the neutral, field-expelling arrangement reads the residue and nothing else does (Section 8);
- that the two directions of the record are the two holographies (Section 5);
- that the bracket's floor is the uncertainty floor made independent of the machine (Section 4.1).

At every other line the located answer is a proposition proved here, cited, or recorded from a companion card.

Inhabitation is not acceptance. That the trace has nine entries with their pairs is well-formedness. The verdicts above are readings of it.

10 What the knower carries

Collected, the proposals of this card come to nine sentences. Equality is a mirror. The run contains its limit. The selected degree is nature's degree. Least work is least action. The carrier is the electron. The sign is what vectors forget. The slack and the spinor are one clock. The device's coupling is the laboratory's. The number under the card is a choice. Nine sentences, nine different debts: a limiting statement is not a physical identification, an action-selection principle is not an equivalence of clocks, and some of them admit finite proofs under explicit assumptions. Each is a calibrated reading in the sense of the numerical card's reading note: it reads a construction, supplies a correspondence, preserves some properties, and stands to be revised by stated evidence.

1. **Equality is a mirror; to solve is to stand where an action is stationary.**
   Reads: Propositions 3.1 and 3.2. *Supplies*: solving $r = 0$ is stationarity of an action with $S' = r$. *Preserves*: roots as stationary points, the reflection as the exchange of roots. *Revised by*: a residual that is no derivative. For a scalar residual there is none; for a vector residual the Jacobian must be symmetric on a simply connected domain, and a residual with nonsymmetric Jacobian refutes the general form by a finite computation.

2. **The run contains its limit.**
   Reads: the bracket and its slip (Section 4.1). *Supplies*: a finite run for a limit. *Preserves*: the walls. *Revised by*: walls that fail to nest. Nested walls with widths tending to zero enclose one point, by the completeness of the reals; completeness is the declaration no finite record supplies.

3. **The selected degree is nature's degree.**
   Reads: the degree hierarchy (Section 3.2). *Supplies*: degree at most three for the fundamental actions. *Preserves*: $D^4 S = 0$ in the declared fields. *Revised by*: a needed action with no cubic presentation in independent fields. The count depends on the presentation (the numerical card's non-Abelian remark), so the evidence must name its fields.

4. **Least work is least action.**
   Reads: Propositions 1.12 and 3.2. *Supplies*: steps of writing for action. *Preserves*: additivity over entries. *Revised by*: a measured relation between logical steps and energy at a named boundary, and its failure to be universal across machines.

5. **The carrier is the electron.**
   Reads: Definition 7.1. *Supplies*: the defined object for the physical electron. *Preserves*: the transformation law, the observation map and the stationary system, which are the standard ones. *Revised by*: any observation of the electron that its stationary system with some coupling does not reproduce at the classical level; the quantum level needs the quantization the card does not supply.

6. **The sign is what vectors forget.**
   Reads: Proposition 3.11. *Supplies*: the coin for the kernel $\{I, -I\}$. *Preserves*: $O \circ U = R \circ O$. *Revised by*: a description by vectors alone that reproduces the 4π fringe of Proposition 8.1. None has; the fringe is observed [21, 24].

7. **The slack and the spinor are one clock.**
   Reads: Proposition 3.12. *Supplies*: one count for two. *Preserves*: complementarity at four and at eight. *Revised by*: already revised in its strong form, at the ninth quarter-turn, where both readings fail. What remains is the two-count comparison; a map or a dynamics preserving both readings would support more.

8. **The device's coupling is the laboratory's.**
   Reads: Definition 7.1(viii). *Supplies*: the certified reading for the inverse fine-structure constant. *Preserves*: the coupling's place in the law. *Revised by*: measurement, which has revised it: 1.80 parts in ten thousand, excluded at the laboratory's precision (Section 7). The narrow form, one stated arrangement, is revised by the fringe phase δ of that arrangement.

9. **The number under the card is a choice.**
   Reads: Theorem 2.7. *Supplies*: a selection for a representative. *Preserves*: the walls. *Revised by*: a procedure inside the family that produces the value, which the theorem excludes under its hypotheses.

The left-hand side is everything a prefix can show, and on this card it is shown or cited in full: that a mark fits any carrier, that two readers agree exactly when their lengths do, that re-lettering is free and is not a symmetry, that the box has one point, that exclusive or needs both definite signatures, that the residual is even and the action odd, that Newton's error is exactly the acceleration term, that the degree forgets the coupling and the derivative keeps it, that the slack is positive at four and negative at eight, that the spinor flips at four and returns at eight while σ_1 returns at four, that transport is a connection and its commutator is the curvature, that the action's stationary system is Maxwell's and Dirac's, and that the verdict slips at four and not at eight. The last line of the middle card asks whether the mechanism demonstrating its proof looks like an electron. This card answers what a record can answer: the mechanism carries a coin, a phase, a transport, a rule of degree three, a coupling and a count, and no reader of it can tell the carrier from an empty one. The Quarter's epigraph asks, "Can you find the polynomial in the physical model of this device?" That is the mark's question, and the patter never answers it. Whether the carrier is an electron is the knower's, and it is handed over calibrated, so that it is carried as a proposal and not mistaken for a receipt.

The identification

One sentence remains, and it is the first one: $T = F$. In the table it is the arm that compares TRUE with FALSE. It stores False. It does not click. No run of the device reaches it. What it carries is not chosen.

Proposition 10.1 (The weights are counted)**.** *Every arm of the table carries the frame, a tape of five marks. An arm whose stored comparison can click carries the coefficient tape of thirteen as well, so that* $18 = 13 + 5$*, and the arm* $T = F$*, which stores* False*, carries the frame alone. The five is fixed three times, and the three tapes agree mark for mark:*

1. *the frame as built: a fabric of two crossed strands, a fibre of two strands before it and one strand after it,* $2 + 2 + 1 = 5$ *marks, each the one truth;*
2. this way*, the smooth model: asked at which stations of the descent the local model of the carrier stops, the prover names five, from the smooth model through the laboratory to the naotrino;*
3. that way*, the yarn: asked the same of the yarn, it names five again, the yarn standing where the smooth model stood and the other four stations shared.*

The thirteen is fixed the same way. The step that sends the local model along the yarn stops at thirteen stations, the three laboratory stations and ten that only the yarn step reaches, and its tape is the coefficient tape.

Measured: each count is the prover's census of the middle card's own elaboration, and each agreement closes by reflexivity (Appendix A). The weighting rule is the one declaration on this table: the arm whose stored fact cannot click loses the thirteen. It does not assign the five. The five is where both ways stop, and it is what remains on the arm when the thirteen is taken away. The two tapes of five are not reached alike. The frame writes the one truth five times at zero distance, a tare. The census counts it five times at the end of the long path, through the carrier, the stations and the naotrino, a count. They are the same tape, and the arm $T = F$ carries it. Strike it, and see what the machine holds.

Theorem 10.2 ($T = F$)**.** *Let the arm $T = F$ carry weight $w \geq 0$, the other three arms* 18*, so that $M_w = 18\,I + N_w$ with $N_w = \begin{pmatrix} 0 & w \\ 18 & 0 \end{pmatrix}$ and $N_w^2 = 18w\,I$.*

1. *If $w = 0$ the arm is struck. Then $N_0^2 = 0$: the one-way crossing that survives cannot close, the crossing cycle stops, and every closed walk keeps its coin, $\operatorname{tr} M_0^n = 2 \cdot 18^n$. Then $\lambda_+ = 18$ and the reading is $\frac{324}{65} \cdot 18 = 89.723\,076\,9\ldots$, a rational number: every TRUE read as TRUE.*

2. *If $w = 5$, the weight Proposition 10.1 counts, the arm is written. Then $N_5^2 = 90\,I$, the clock runs, $\lambda_+ = 18 + 3\sqrt{10}$, and the reading is $\frac{972(6+\sqrt{10})}{65} = 137.011\,290\,5\ldots$, irrational. The whole of the difference, $47.288\,213\,6\ldots$, is the growth rate of N_5, whose walks cross the arm $T = F$ at every other step.*

3. *Take that reading for the coupling, $e^2/4\pi = 1/137.011\,290\,5\ldots$. Then the chain of Section 6 carries it through Theorem 6.7 and Definition 6.8 to the minimally coupled Maxwell–Dirac stationary system, Proposition 6.9, under the assumptions named there: a Lorentzian background, the spinor representation, the kinetic terms, the mass, and, for quantum electrodynamics proper, quantization and renormalization.*

Proof. N_w is off-diagonal, so $N_w^2 = 18w\,I$ and the eigenvalues of M_w are $18 \pm \sqrt{18w}$. At $w = 0$ the matrix is triangular with both eigenvalues 18; at $w = 5$, $\sqrt{90} = 3\sqrt{10}$. The reading is $\frac{324}{65}\lambda_+$ in both cases (the cheat sheet's Propositions 9.3 and 9.4). Item 3 is Section 6 with the coupling supplied. *Proved here* for items 1 and 2; item 3 as calibrated where each of its steps enters. □

So the number is in the machine exactly when $T = F$ is written. Strike the arm and the machine holds $89.72\ldots$, the device agreeing with itself, a fraction, nobody's constant. Write it, though it never clicks and no run ever reaches it, and the machine holds $137.011\,290\,5\ldots$, and the number carries through transport and curvature into the coupling of Maxwell's and Dirac's equations. That is what $T = F$ buys. It is written on the card. It is not read by the machine. Under the weighting rule it weighs in the count all the same, and the theorem shows what it contributes. What that counted structure is taken to represent, a number in the machine that is the coupling of an electron, so that every other TRUE stands for FALSE, is the identification, and it is the knower's. *Proposed.* The laboratory has measured the coupling the identification would deliver, $137.035\,999\,177(21)$, and the delivery is short by 1.80 parts in ten thousand (Section 7).

For the reading that makes the identification: TRUE equals FALSE. Write it, and there is a number in the machine. Read it, and there is electrodynamics. The card has not lifted. The knower who reads it so has.

Weird that if you believe there is a number in there you can derive QED.

Isn't that a daisy?

A Sources

This appendix is the only part of the card that names the middle card's files and declarations. The body refers to it only as "the middle card." Paths are relative to `CARD_2/Measurement/`. Declarations are cited by name and not by line; the working tree was inspected on 29 October 2026, and again on 3 November 2026 for the carrier encoding, the readers and the closed walks; nothing here reports a build. Companion cards are cited by printed result number: the geometric card (*The Frinkahedron*, build of 22 October 2026), the numerical card (*Four Models and a Common Fourth-Order Closure*, 29 October 2026), the Quarter, the bent card (*The Infinite Loop Considered Harmless*) and the cheat sheet (*A Knower's Guide to Orbital Mechanics*).

The carrier's encoding. `Episode01.lean` declares `Number` with constructors `zero : FACT → Number` and `one : FACT → Number → Number`, and `CarrierProcess` with fields `symbol : FACT`, `quarter : Number` and `interact_with_the_universe : Number → Number`, the last with a default value. A `Number` is a nonempty record: `Number.peano` (`Episode08.lean`) sends `zero f` to the one-symbol record `[f]` and `one f n` to `f` followed by the record of `n`; on nonempty records `tapeNumber` (`Episode12.lean`) inverts it. Under this map the continuation of Definition 1.2 is `one f n` $\mapsto$ `n`; the slot q is `quarter`; the symbol s is `symbol`; and ι is the default `interact_with_the_universe`, which sends `zero _` to `zero FACT.Truth` (clause (ii)) and `one g _` to `one g quarter` (clause (i)). Its match inspects only the constructor of its argument and never `quarter`: carried, not queried. The structure permits a carrier to override the default; the definition is of the default, which is the only interaction the inspected tree uses. The executable slot is therefore a `Number`, and the record of the definition is its image under `Number.peano`.

This card	Companion card	Middle card (declaration)
Def. 1.2, carrier	Quarter §6 (affixing)	`Episode01.lean`: `CarrierProcess` with fields `symbol`, `quarter`, `interact_with_the_universe`; `Number`; `Episode08.lean`: `Number.peano`; `Episode12.lean`: `tapeNumber`
Def. mark; click	cheat sheet §6	`Episode01.lean`: `FACT` (`truth`, `click`), `FACT.Truth`; `Episode12.lean`: `the_click`
Def. 1.4, three readers	bent card Props. 1.3, 1.4	`Episode08.lean`: `naotrino_detector` (the register), `List.length` (the tally); `Episode12.lean`: `the_click_count`
Prop. 1.5	Quarter §6 (two counting layers)	`Episode08.lean`: `adding_in_the_fencepost_gives_a_monomial`; `Episode12.lean`: `the_clicks_agree`, `the_clicks_read_the_number`

This card	Companion card	Middle card (declaration)
Def. 1.6, Prop. 1.7	—	`Episode12.lean`: `the_click_count` (a payload reader); `read_past_quarter`, `read_past_quarter_is_length` (a reader in the length family)
Prop. 1.8	cheat sheet Prop. 6.1	`Episode12.lean`: `the_card_between`, `the_mark_between`, `the_tape_between_deals_rung01`
Prop. 1.9	cheat sheet Prop. 6.2	`Episode12.lean`: `ComputerProgram.ofTape`, `read_ofTape`, `the_other_machine_agrees`
Props. 1.10, 1.11	numerical card §7.4	`Episode12.lean`: `relettered`, `relettering_is_free`, `any_count_of_the_lengths`
Prop. 1.12	numerical card Rem. 8.3	`Episode11.lean`: `TYPESET.universe_id`
Prop. 2.1	numerical card §7.5	`Episode08.lean`: `the_rational_idea_of_relativity`; `Episode12.lean`: `the_box_has_one_point`, `the_box_reads`
Prop. 2.2	cheat sheet Prop. 6.3	`Episode05.lean`: `TruthOrder`, `FACT.SAME`; `Episode08.lean`: `a_partitioning_of_quotients`
Def. 2.4	geometric card Def. 4.1, Props. 4.2, 4.3	`Episode08.lean`: `theReadingCertified`
Thm. 2.7	Quarter Thm. 3.1	the axiom sweep reported in the Quarter; `Episode12.lean` `#print axioms` lines
Prop. 2.10	numerical card §3.5 (multiplier vs. penalty)	`Episode12.lean`: `Signature`, `Signature.sign`, `xorClauses`, `xorRow`, `xorTarget`, `step_correspondence`
Prop. 2.11	cheat sheet Prop. 6.4	`Episode12.lean`: `SAME_DIFFERENCE`, `same_difference_clicks`, `the_click_diagonal`
§3.1	cheat sheet Props. 3.1, 3.2; numerical card Prop. 6.1	`Episode12.lean`: `the_acceleration_is_constant`, `the_first_derivative_squared_is_the_discriminant` (stated there with the middle card's coefficients; here without)

This card	Companion card	Middle card (declaration)
§3.2, Props. 3.7, 3.8	numerical card Thms. 1.1, 2.4, Prop. 2.5, §2.5	`Episode04.lean`: `Variation` with constructors `the_newton`, `the_gateaux`, `the_frechet`; `Episode12.lean`: `gateauxAt`, `frechetAt`, `theConvectiveGap` and its margin
Prop. 3.9	cheat sheet Prop. 4.1; bent card §4	`Episode08.lean`: `the_slack`, `once_around_the_integral`, `twice_around_the_integral`, and the three theorems that read its sign (the first time around, through the bent card, the second time around)
Props. 3.10, 3.11	cheat sheet Prop. 7.1	`Episode05.lean`: `UNIVERSAL.once_around`, `UNIVERSAL.twice_around`; `Episode01.lean` margin on the clock and clock complement
Prop. 3.13	numerical card Rem. E.1	`Episode08.lean`: the register's all-ones identities
Props. 4.1, 4.2, 4.3	geometric card §3.4, Props. 3.3, 3.4, §6.3	`Episode03.lean`: `Equivalation` and `DigitalProcess.tick` (the exclusion arm); `Episode02.lean` margin on the six-state sensor
Prop. 4.4	Quarter §6	`Episode12.lean`: `the_card_between` dealt through `deal_rung`; the scratch comparison of all steps is reported in the Quarter and not here
§4.1	Quarter §6 (two floors)	`Episode05.lean`: `Closure.same`, `Closure.different`; `Episode12.lean`: `theCourant`, `bracketed_at_the_slip`; `Calibration/LeanCalibration.lean`: `register_noise_floor`
§5, Def. 5.1	both cards §8; numerical card Rem. E.2	`Episode10.lean`: `REVIEWED`; `Episode11.lean`: `TYPESET`, `PressVerdict`, `pressCheck`

This card	Companion card	Middle card (declaration)
§6, Def. 6.3 (the closed walks)	numerical card §2.4	`Episode12.lean`: `the_closed_words`, `the_closed_walks`, `the_closed_words_are_closed_walks`, `the_crossings`, `every_closed_walk_crosses_evenly`, `the_winding`, `the_turning_walks`, `every_closed_walk_turns_evenly`; no declaration of a connection, a transport between points, or a curvature
Prop. 10.1, Thm. 10.2	cheat sheet §9	`Episode08.lean`: `the_coordinate_system`, `the_dimensions_of_universe`, `the_coordinate_system_gauss_length`; `episode7_elaboration_of` `this_way` as `the_data_table`, `that_way` as `the_train_schedule`, `the_calculus` as `the_differential_form`; `the_data_table_is_the_gauss`, `the_train_schedule_is_the_gauss`, `the_differential_form_is_the_coefficient`, `bent_card`, `the_macguffin`; `Episode12.lean`: `the_arm_stores_a_fact_that_can_click`, `the_arm_weight`, `the_click_table_is_derived`
Def. 7.1	cheat sheet §6, §7; numerical card Ch. 6	`Episode07.lean` margin listing the electron's values; `Episode04.lean`: `Knowledge.jarjar` and its margin casting the part; `Episode12.lean`: `readers_interpretation_of_the_experience`; for (viii), `what_would_you_call_this_number?`
Prop. 7.3	numerical card §2.4	`Episode01.lean` foreword on the two readings of Q.E.D.
§7.1	Quarter §6 (the horizon)	`Episode08.lean`: `the_photon_worldline` and the three encodings that follow it (position, velocity, acceleration, each zero)
§8	cheat sheet §6.2, §7; numerical card §6.7	`Episode05.lean`: `AtreyuProcess` margin; `Episode07.lean`: `YARN_THEORY` margin
§9	both cards §8.3	`Episode11.lean`: `pressCheck`

B The reader's experiment

Every result calibrated *proved here* is a written record: hypotheses, a sequence of displayed steps, and a conclusion. Read as an experiment it has a setup, an apparatus (whoever checks the steps), a law (the rule each step obeys), a procedure (the order of the steps), observations (each step seen to hold) and a reading (the conclusion). Four runs on paper reproduce the results on which the press check and the transport rest.

Run 1: the slack. Write the odd numbers below eight and below sixteen; form their least common multiples, 105 and 45045; divide each by each odd number and subtract the quotients from the multiple. Observe 34 and -982. Each line is an integer division and a subtraction. This is Proposition 3.9.

Run 2: the signatures. Write the two clauses of Proposition 2.10 over four slots. For each of the four signatures, sign the slots, read each literal, and evaluate the formula on the four assignments. Observe the four rows of the table. Conjoin the first two rows and observe $[F, T, T, F]$; conjoin the last two and observe $[F, F, F, F]$. If any single row equals $[F, T, T, F]$, Proposition 2.10 is wrong.

Run 3: the eighth root and the rotating observable. For $k = 0, \dots, 8$ and $\theta_k = k\pi/2$, write $U(\theta_k) = \mathrm{diag}(e^{-ik\pi/4}, e^{ik\pi/4})$. Record where $U(\theta_k) = I$: at $k = 0$ and 8. Compute $U(\theta_k)\sigma_1 U(\theta_k)^*$, whose upper-right entry is $e^{-ik\pi/2}$, and record where it equals σ_1: at $k = 0$, 4 and 8. Then compute $U(\theta_k)\sigma_3 U(\theta_k)^*$ and observe σ_3 at every k: a diagonal observable is the control, and it cannot show the pattern. These are Propositions 3.10 and 3.11, and the trace of Section 9 is their rows. If σ_1 returns at $k = 2$, the trace is wrong.

Run 4: the commutator. Write $D_\mu = \partial_\mu + ieA_\mu$. Expand $D_\mu D_\nu \psi$ into five terms; exchange μ and ν; subtract. Observe that four terms cancel in pairs and that $ie(\partial_\mu A_\nu - \partial_\nu A_\mu)\psi$ remains. Then replace ψ by $e^{-ie\chi}\psi$ and A_μ by $A_\mu + \partial_\mu \chi$ in $D_\mu \psi$ and observe the factor $e^{-ie\chi}$ come out whole. This is Proposition 6.5(c) and Theorem 6.7.

A second run by another reader is a repeat; its readings either equal the first or do not, and the comparison is the press check. An exact replay of the first run's record, line by line, observes nothing. The record supplies the procedure; the experience, every observation in it, happens in the one who runs it.

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